Cellular Biology

Dr. Thanh Huong LE



Cellular Biology

Dr. Thanh Huong LE



Assessment

	Percentage	Type
Attendance/Attitude	10%	
Assignment	00%	
Mid-term exam	20%	Writing exam/ Multi-Choice Question
Practical	20%	Attitude and report
Final exam	50%	Writing exam/ Multi-Choice Question

Class	Contents	Hours		
		Lect.	Exr.	Prc.
1	[1] Topic 1: The fundamental unit of life	03		
2	[2] Topic 2: The chemical components of the cells	03		
3	[3] Topic 3: Eukaryotic cell structure	12		
4	[4] Topic 6: Cell cycle and cell communication	03		
5	[5] Topic 8: Apoptosis, autophagy, cancer, and stem cell	03		
6	[6] Topic 7: Cell metabolism	02		
7	[7] Topic 9: Prokaryotic cell and virus	02		
8	[8] Topic 10: Cell techniques and their applications	02		
9	[9] Topic 11: Lab work			06
10	Tutor		05	
	Total:	**30**	**05**	**06**

Cellular Biology

Chapter 1. The fundamental unit of life

Dr. Thanh Huong LE

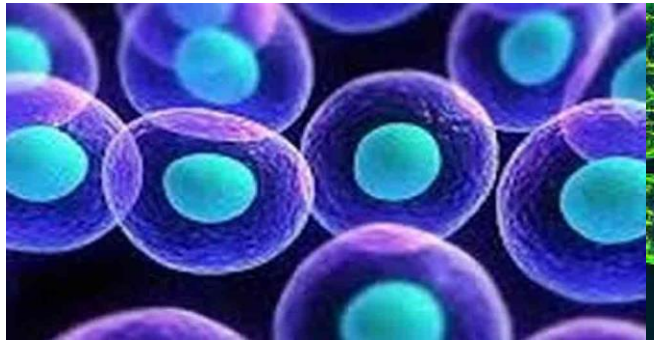


Essential concepts

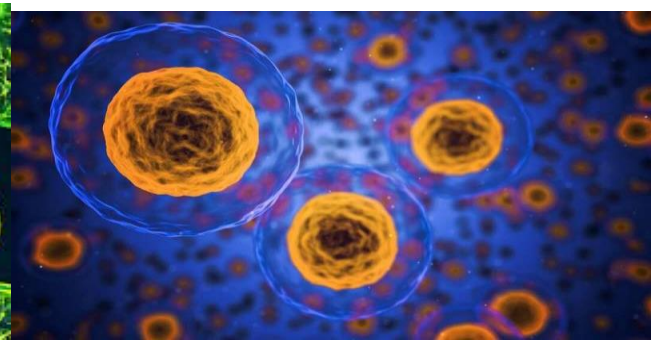
- Give the definition of the cells
- Difference between Prokaryotes vs Eukaryotes cells
- Describe domains and kingdoms

Part 1

What is a cell?

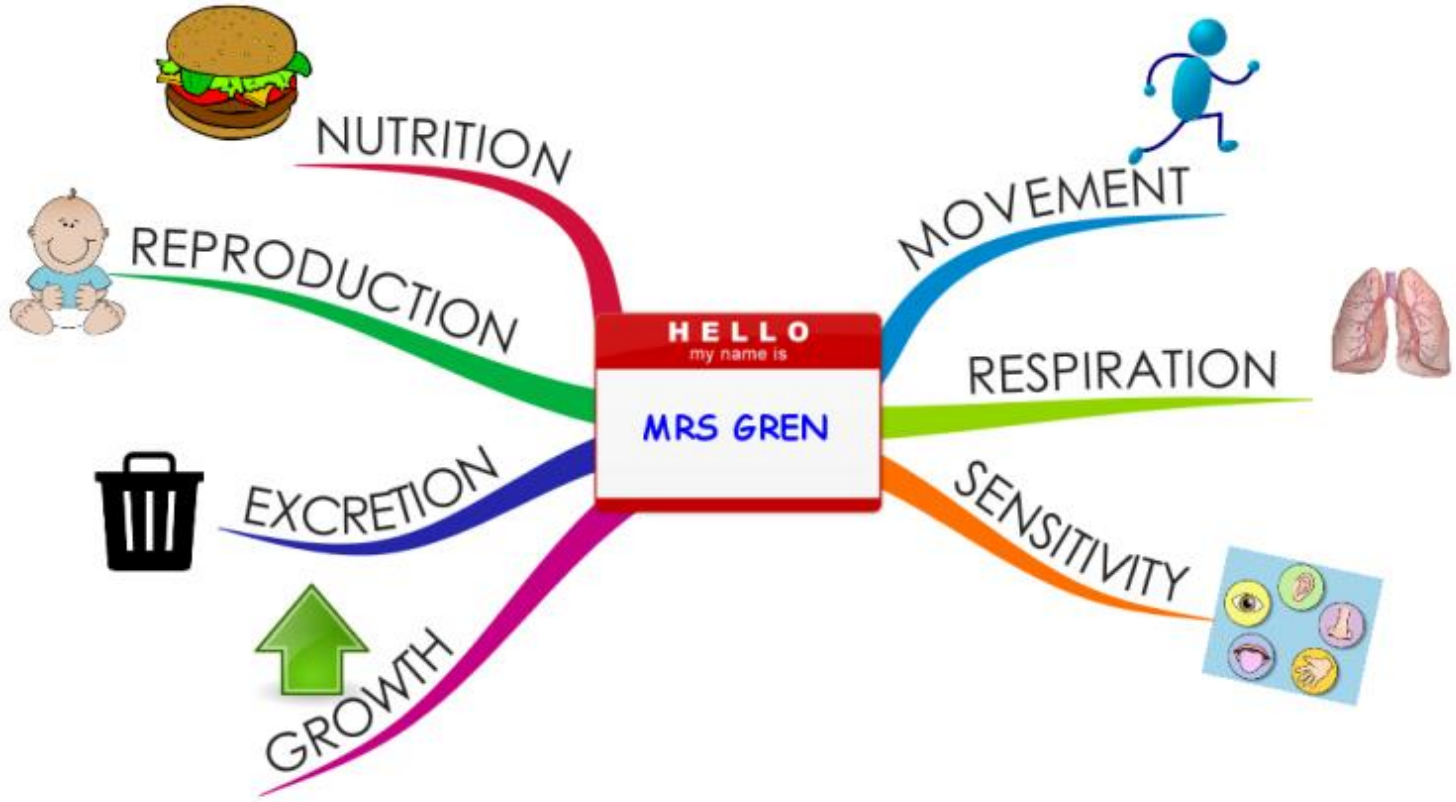






1. Characteristics of living organisms

All living organisms have a set of seven characteristics that separate them from non-living objects.



All living organisms are constructed from cells.







CELL







organism



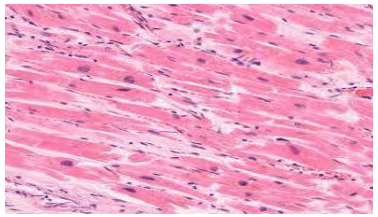


organs

tissues



cells



Fundamental basic unit of life

Unity and diversity of Cells

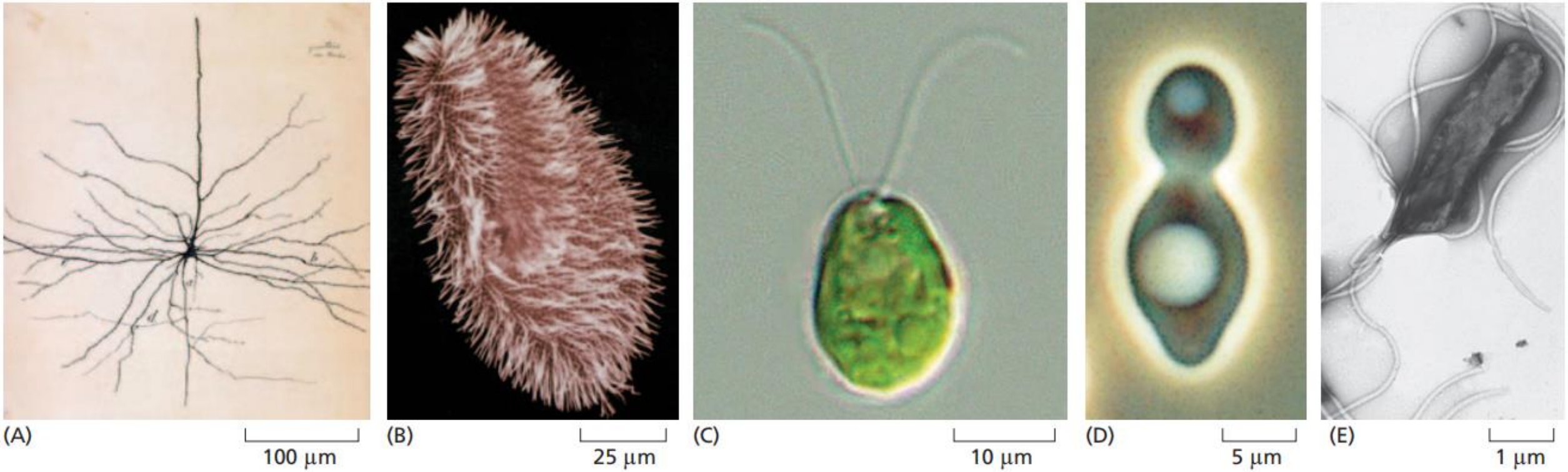
UNICELLULAR ORGANISMS



Single celled organisms

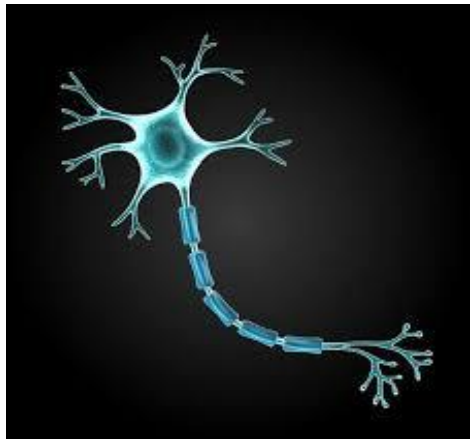
MULTICELLULAR ORGANISMS





Cells come in a variety of shapes and sizes

(A) Drawing of a single nerve cell from a mammalian brain
(B) *Paramecium.* This protozoan—a single giant cell—swims by means of the beating cilia that cover its surface.
(C) *Chlamydomonas*. This type of single-celled green algae is found all over the world
(D) *Saccharomyces cerevisiae*. this yeast cell, used in baking bread, reproduces itself by a process called budding.
(E) *Helicobacter pylori*. This bacterium—a causative agent of stomach ulcers—uses a handful of whip like flagella to propel itself through the stomach lining.



Neuron



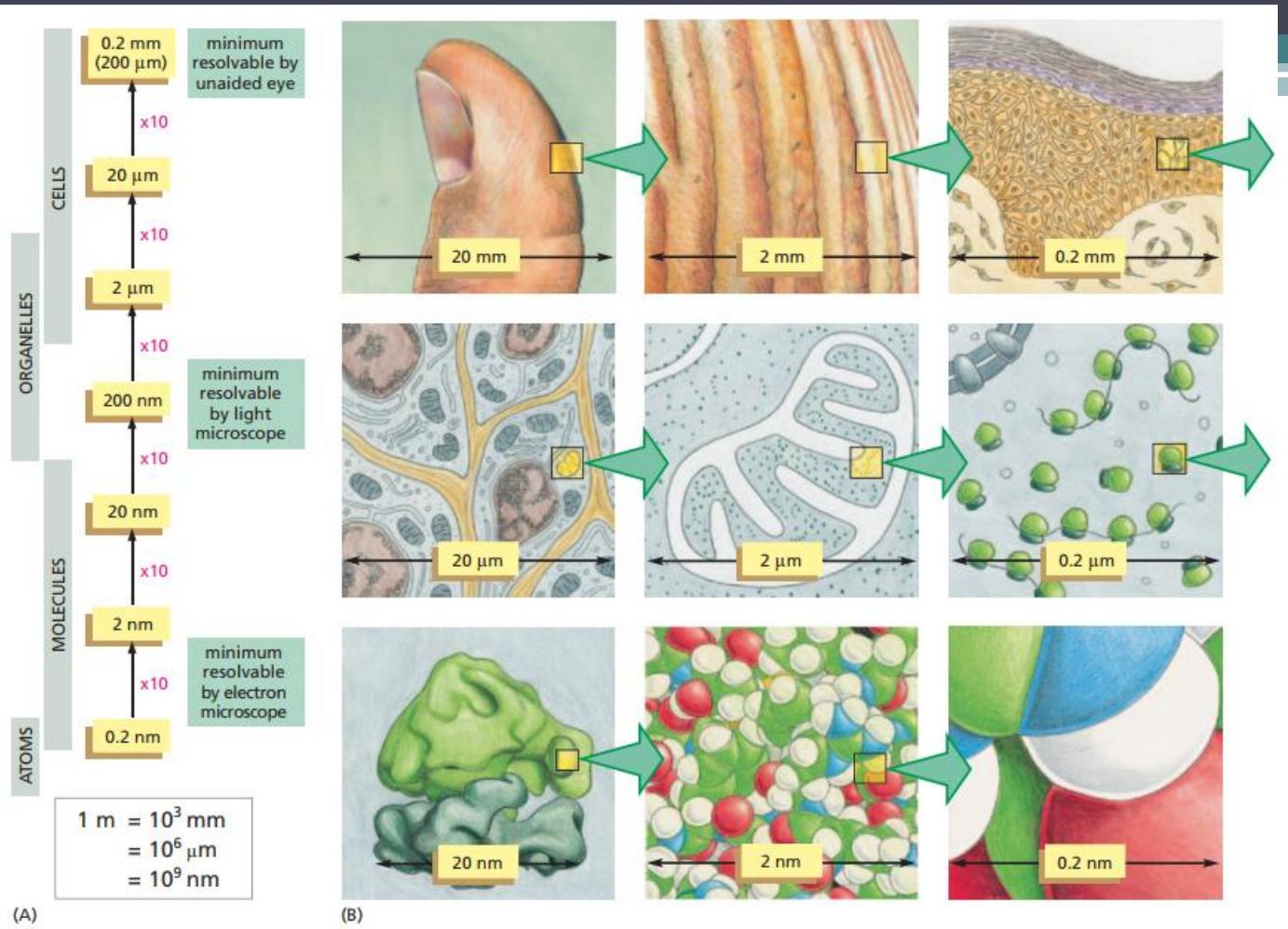
RBC



Sperm

Difference in:

- **Size**
- **Shape**
- **Structure**
- **Function**



How big is a cell and its components?

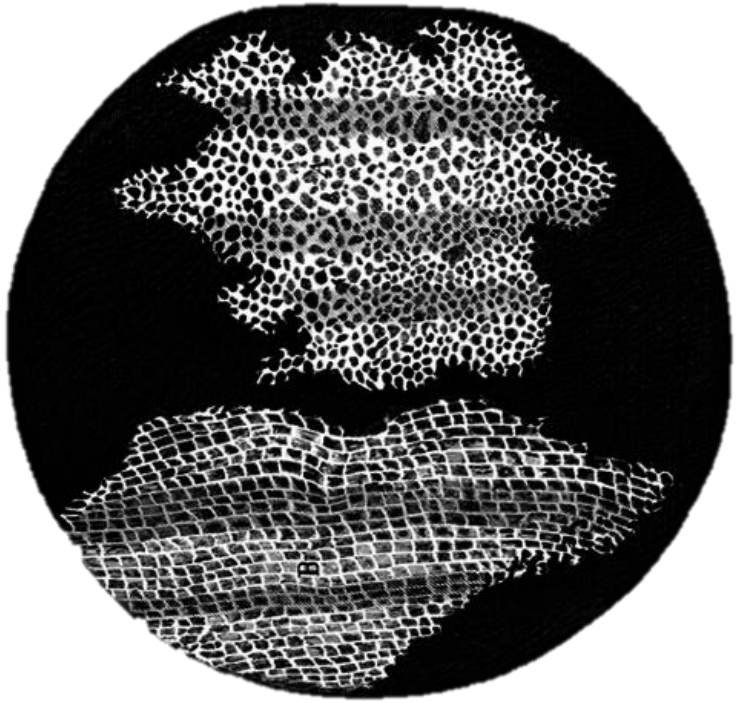
(A) the sizes of cells and of their component parts, plus the units in which they are measured.

(B) Drawings to convey a sense of scale between living cells and atoms. Each panel shows an image that is magnified by a factor of 10 compared to its predecessor—producing an imaginary progression from a thumb, to skin, to skin cells, to a mitochondrion, to a ribosome, and ultimately to a cluster of atoms forming part of one of the many protein molecules in our bodies. Note that ribosomes are present inside mitochondria (as shown here), as well as in the cytoplasm.

Cells Under the Microscope



Robert C. Hooke (1635-1703)



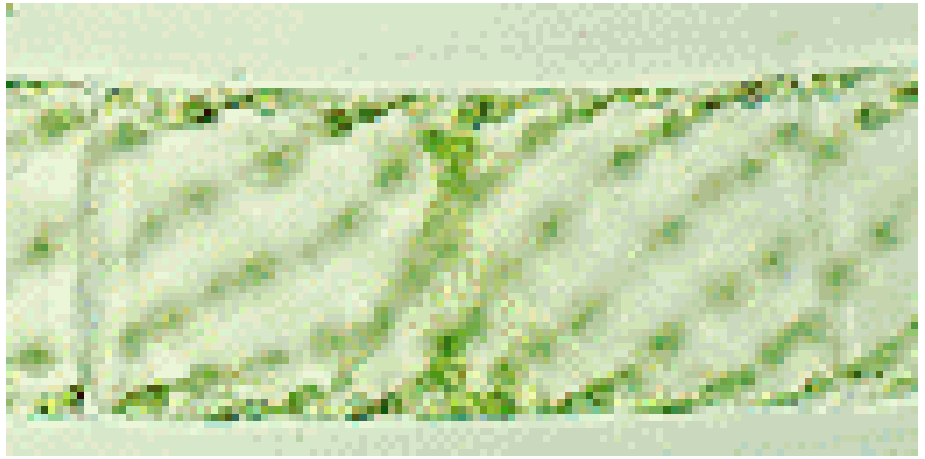
1665: The cell was first discovered and named by Robert Hooke

Hooke viewed a thin cutting of cork and discovered empty spaces contained by walls which he termed cells

Read more at https://www.science-of-aging.com/timelines/hooke-history-cell discovery.php#4YZWECUjTtl3kZlh.99



Antony van Leeuwenhoek (1632-1723)



1674 – Leeuwenhoek discovered protozoa



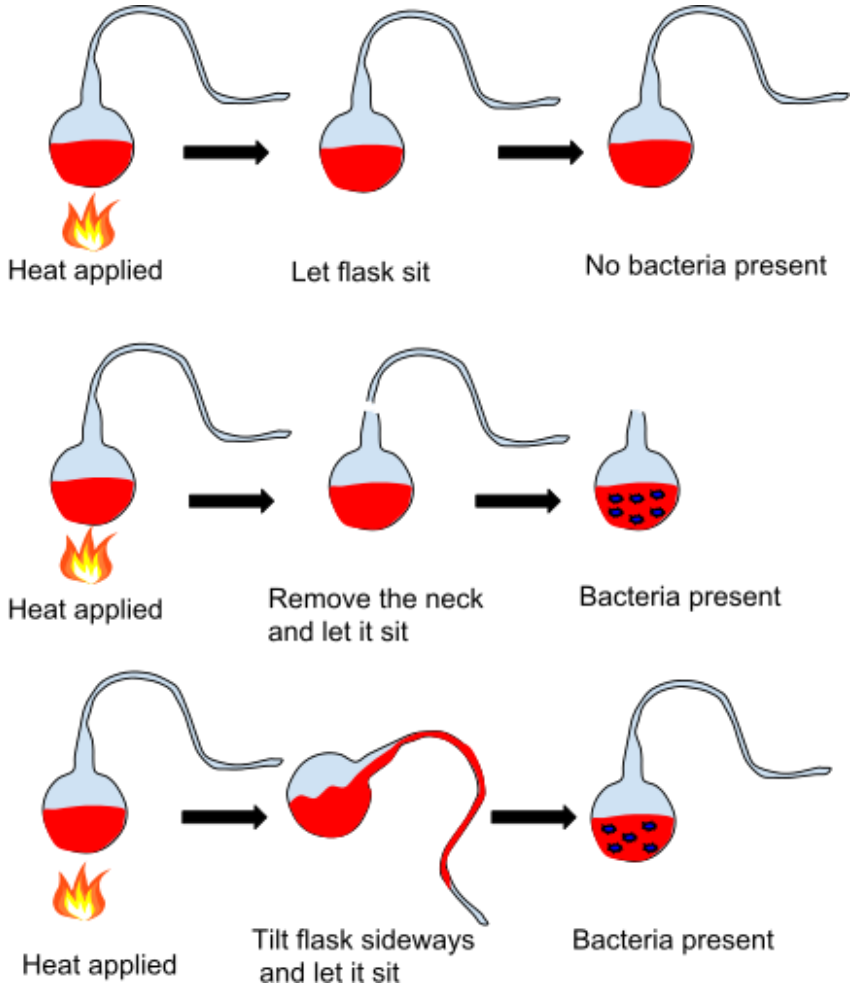
Gave descriptions of many protists, including this ciliate
He saw bacteria some 9 years later,

1838 – Schleiden and Schwann proposed cell theory

1. The cell is the unit of structure, physiology, and organization in living things.

2. The cell retains a dual existence as a distinct entity and a building block in the construction of organisms.

3. Cells form by free-cell formation, similar to the formation of crystals (**spontaneous generation**).



Pasteur's test of spontaneous generation (1960s)

By sterilizing a food source and keeping it isolated from the outside, Pasteur observed no putrefaction of the food source (top panel).

Modern Cell Theory

1. All known living things are made up of cells.

2. The cell is structural & functional unit of all living things.

3. All cells come from pre-existing cells by division. (Spontaneous Generation does not occur).

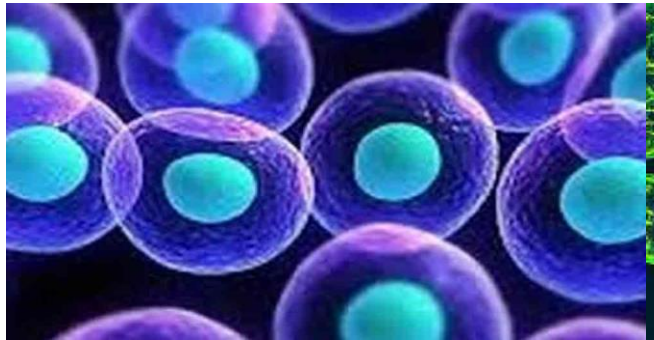
4. Cells contains hereditary information which is passed from cell to cell during cell division.

5. All cells are basically the same in chemical composition.

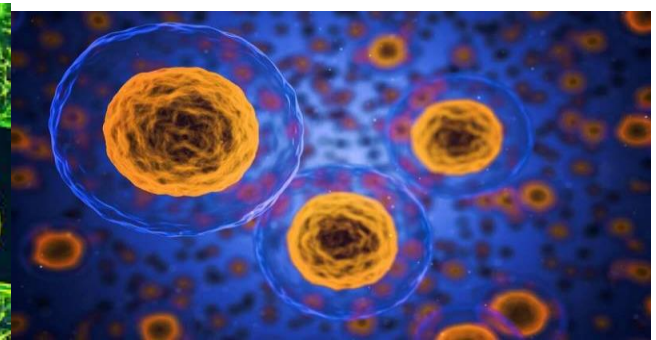
6. All energy flow (metabolism & biochemistry) of life occurs within cells.

Part 2

Domain of life

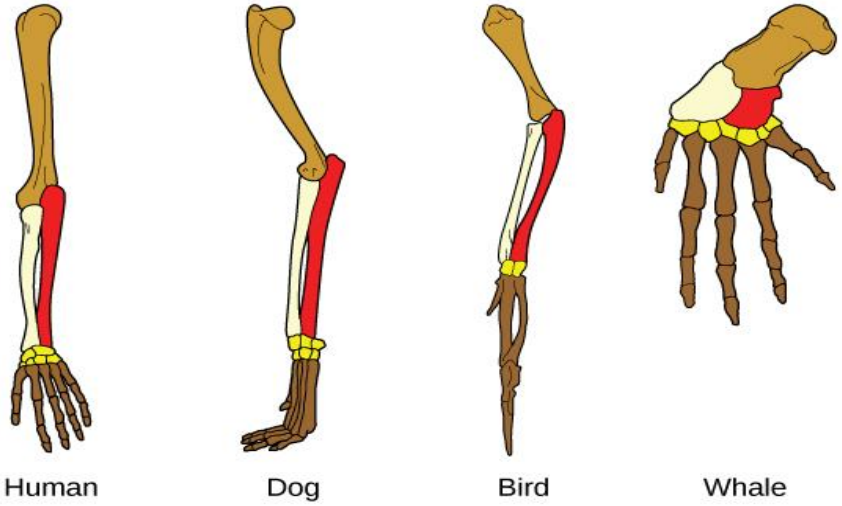






How organisms are classified

- Organisms can be classified by the features that they share.
- Classification systems aim to reflect evolutionary relationships (change of adaptive features of a population over time, as a result of natural selection)
- **Morphology**: the overall form and shape of their bodies: For example wings, legs…
- **Anatomy**: the detailed body structure determined by dissection





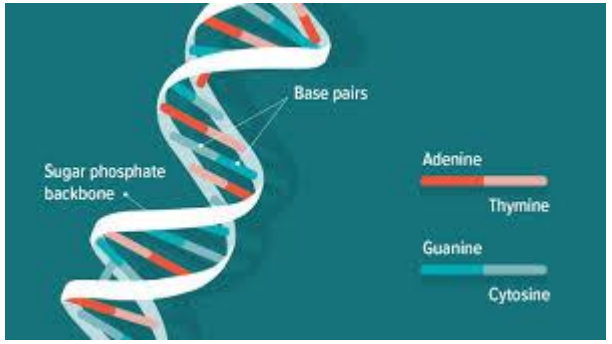
DNA for classification:

- DNA sequencing of different species helped to classify organisms using a more scientific approach
- The more similar the base sequences in the DNA of two species, the more closely related those two species are (and the more recent their common ancestor is)

Important of classification:

- Conservation
- Evolutionary relationship







Classification concepts:

- Species: a group of organisms that can reproduce to produce fertile offspring

The binomial system

A system of naming species, in which the scientific name of an organism is made up of two parts showing the genus and species

Normal name: Human
Scientific name: *Homo sapiens*



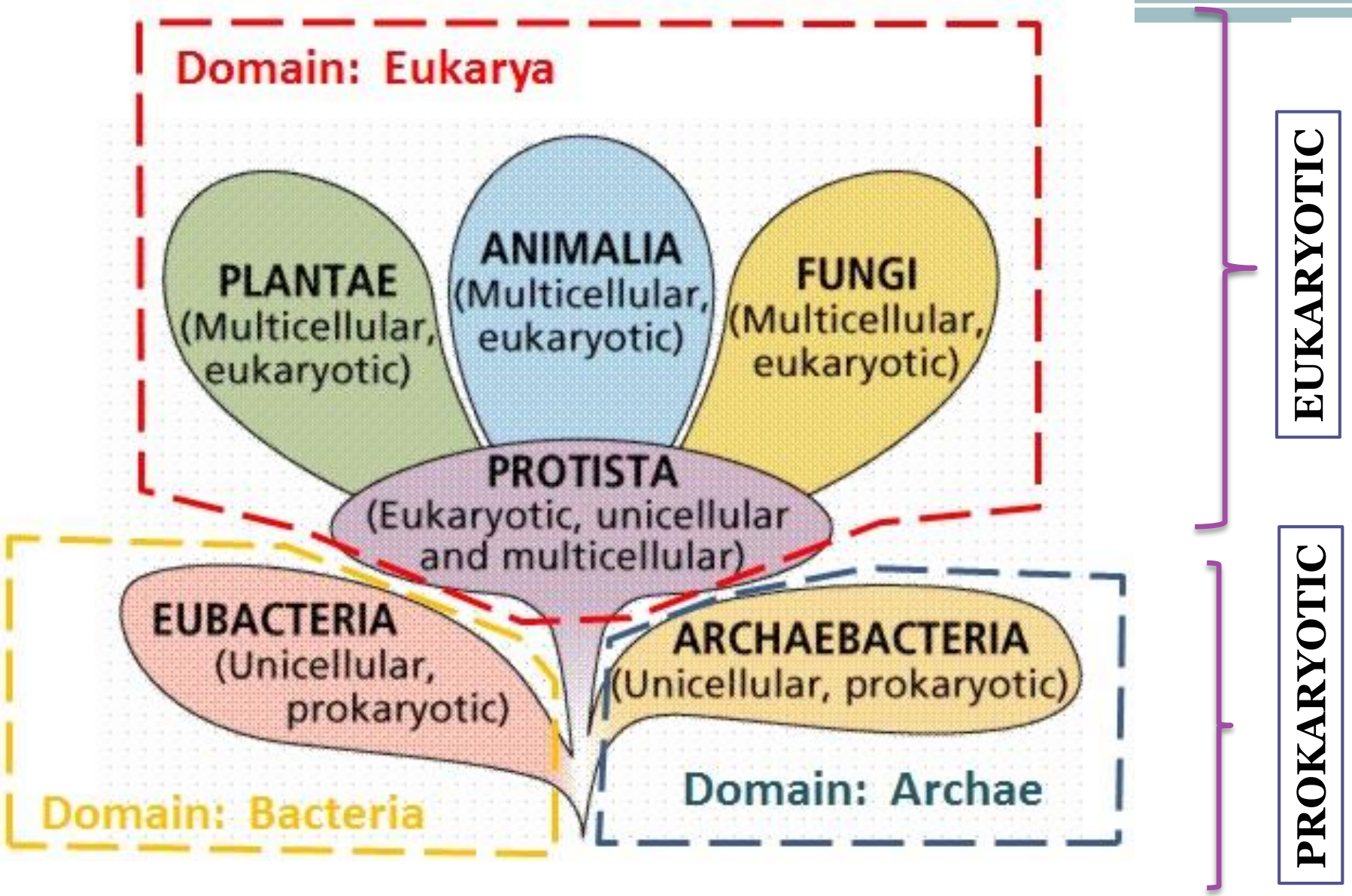


Image from Purves et al.,

Domain Bacteria

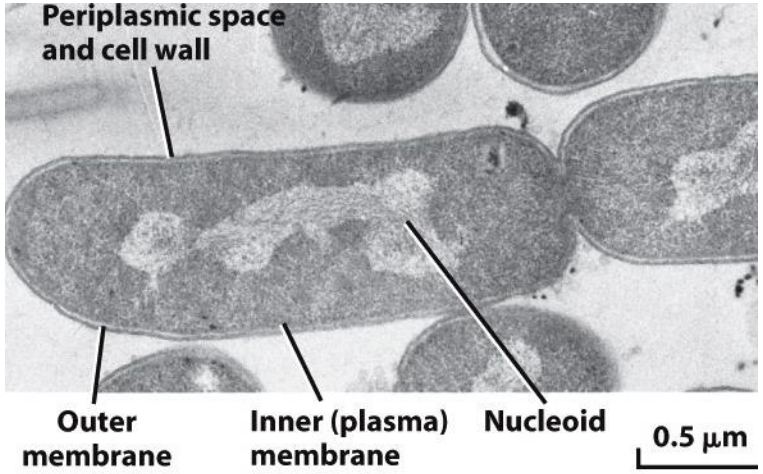
- Unicellular, although a few live colonial lives at least some of the time (e.g. Cyanobacteria)

- Include: **autotrophic** and **heterotrophic**

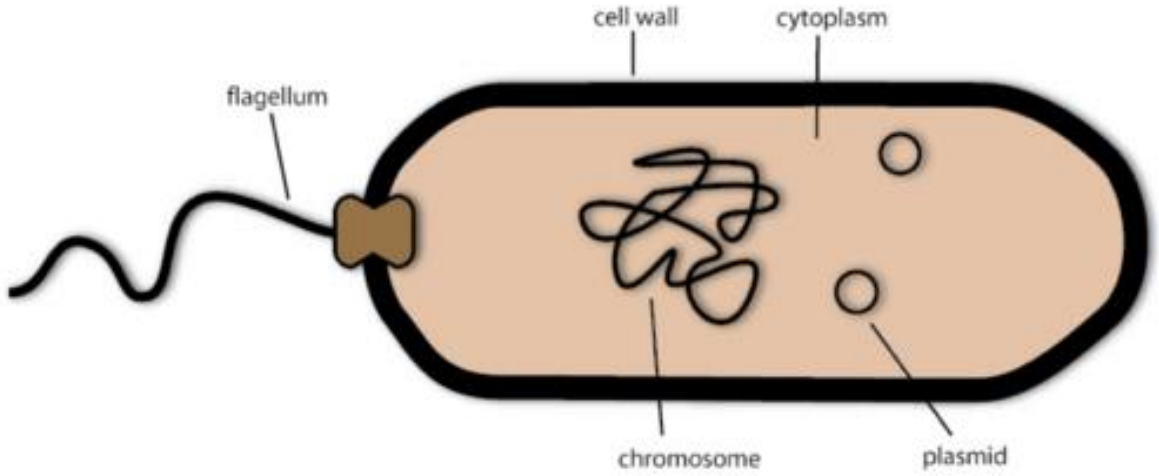
- They are small individually, but make a huge part of the earth's biomass

Ex: a single *E.coli* bacteria has a dry weight of $25x10^{-14}$ g, but total account for 1-1.5 kg of human's weight





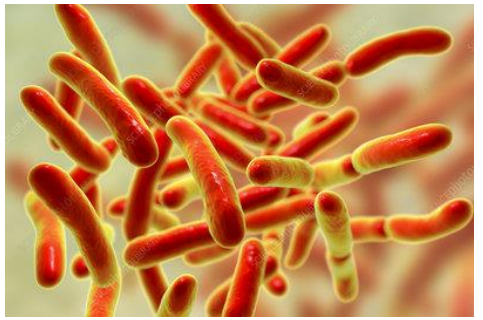
Electron micrograph of a thin section of *Escherichia coli* a common intestinal bacterium



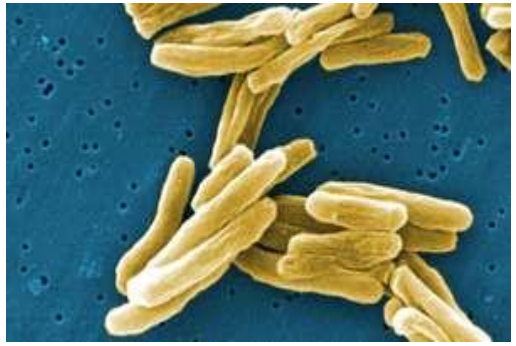
Simplest structure of bacteria

Adapted for growth under many diverse conditions

Temperatures:

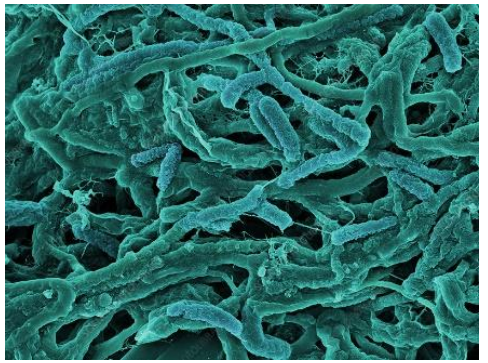


❖ Most known prokaryotic species, called mesophiles grow best between 25° C and 40°C



❖ Psychrophiles grow best at temperatures between 15°C and 20°C, but some species can live at 0° C

❖ Thermophiles grow at temperatures between 50°C and 60°C, but some can tolerate temperatures as high as 110°C



Domain Archaea

Unicellular, prokaryotic

❖ **Lack nuclei** are similar in size and shape(s), which initially suggested that they were closely related to bacteria and were therefore prokaryotes.

❖ rRNA gene sequence comparisons: **archaea** were more **closely related to eukaryotes than bacteria.**

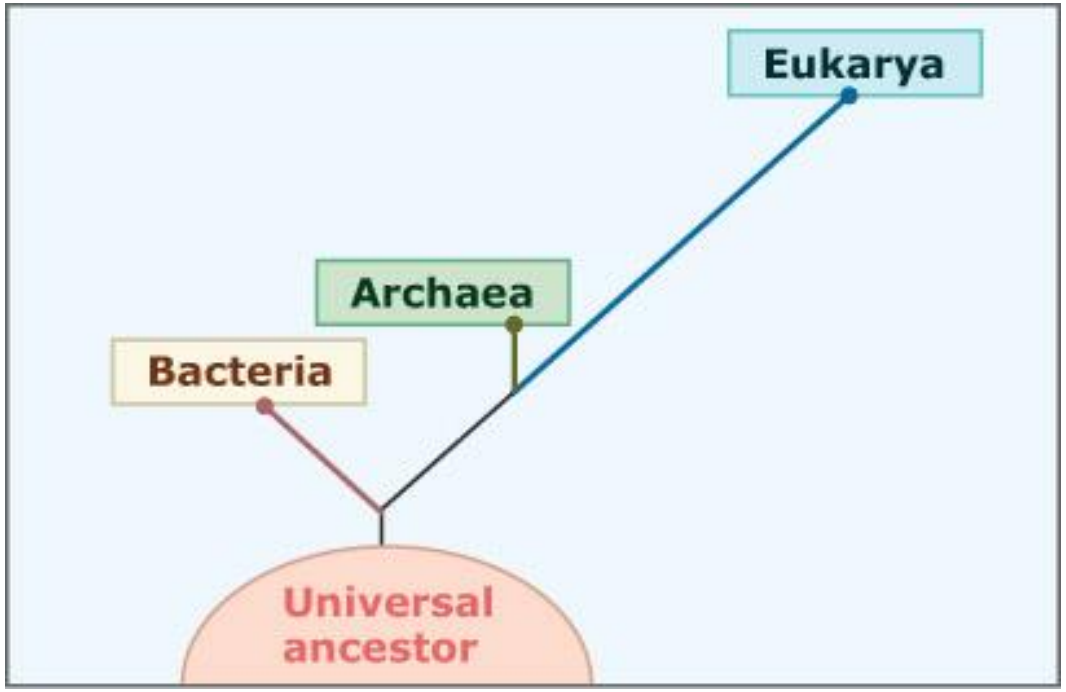


Figure 01.F07: Phylogenetic analysis by molecular methods shows that organisms can be classified into three domains.

Domain Archaea

❖The **archaea** include the most diverse **extremophiles** :

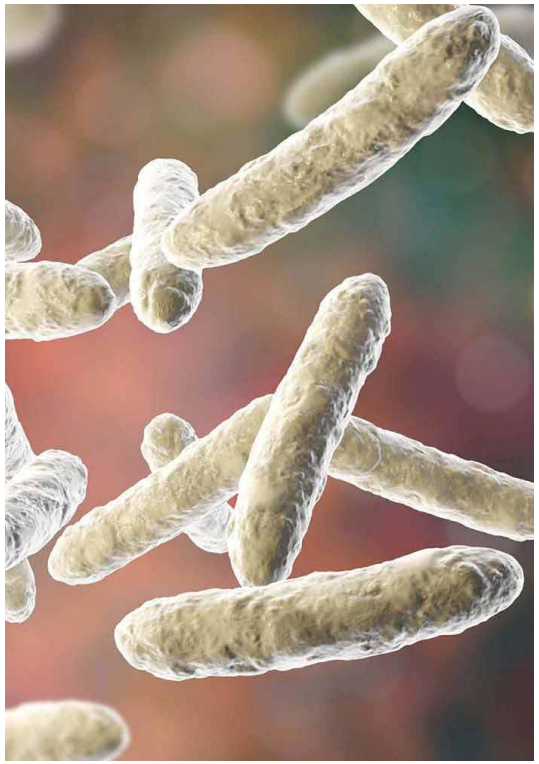
Acidophiles: grow at acidic (low) pH.

Alkaliphiles: grow at high pH.

Toxicolerants: grow in the presence of high levels of damaging elements (e.g., pools of benzene, nuclear waste).

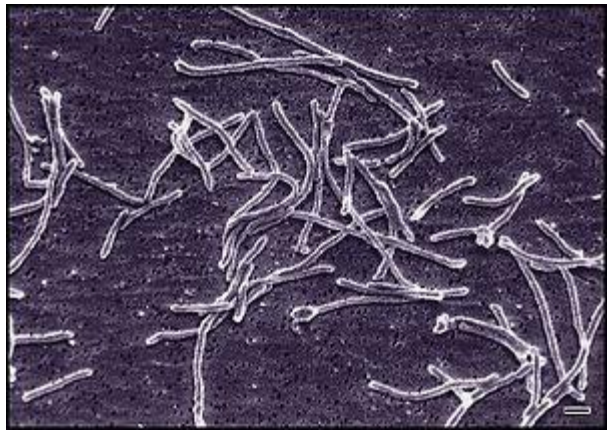
Psychrophiles: grow best at temperatures 15°C or lower

Thermophiles/hyperthermophiles: organisms that grow best at 40°C or higher, or 80°C or higher, respectively. *Pyrolobus fumarii*, shown below, can live at a temperature 113°C



Application

Taq polymerase, the heat-stable DNA polymerase that made the *polymerase chain reaction* (PCR): *Thermus aquaticus*



Thermus aquaticus

Best growth at 65 to 70°C but can survive at temperatures of 50°C to 80°C



Pyrococcus furiosus

It grows between 70°C and 103°C, with an optimum temperature of 100°C

***Pfu* DNA polymerase**

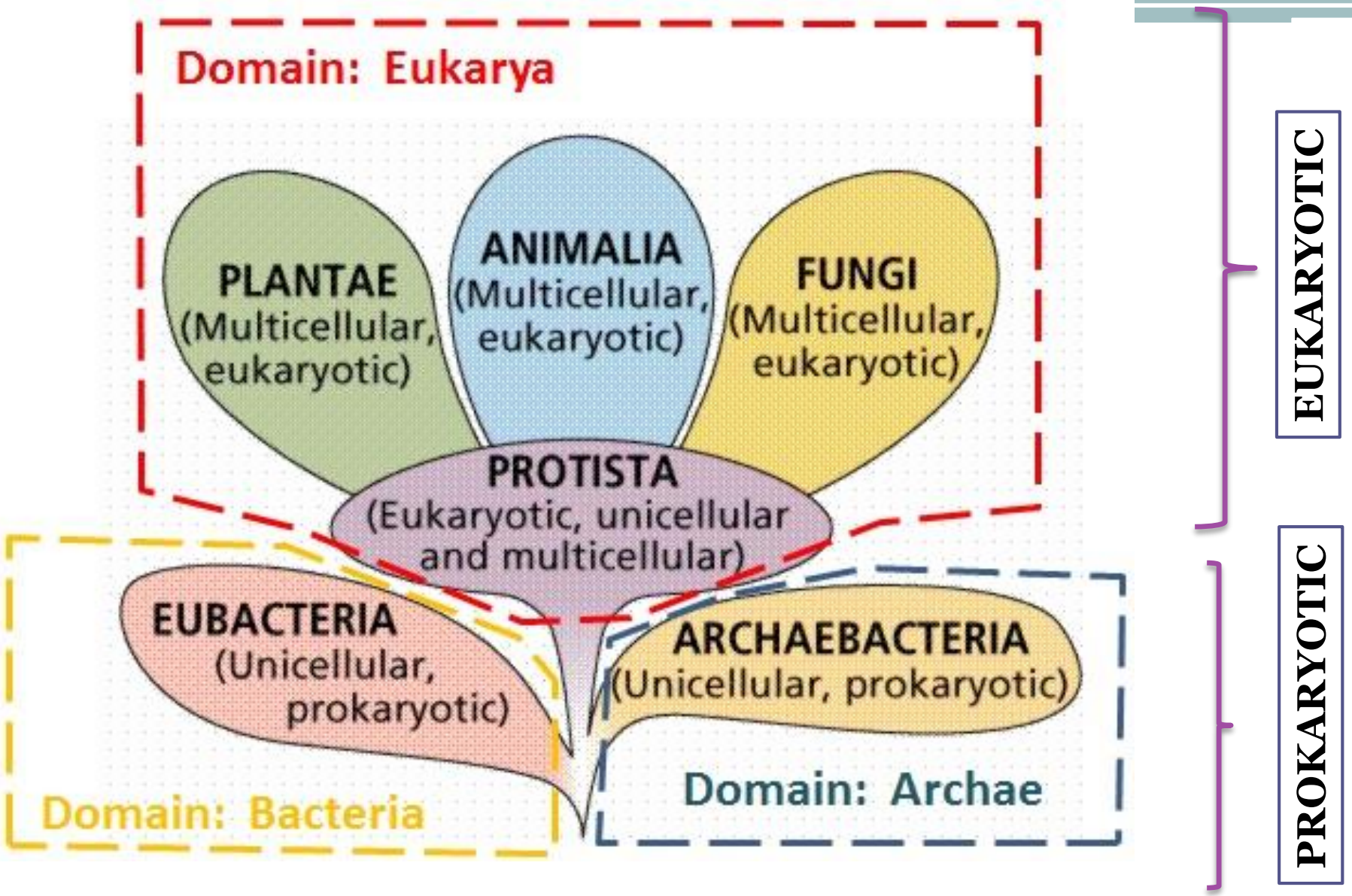
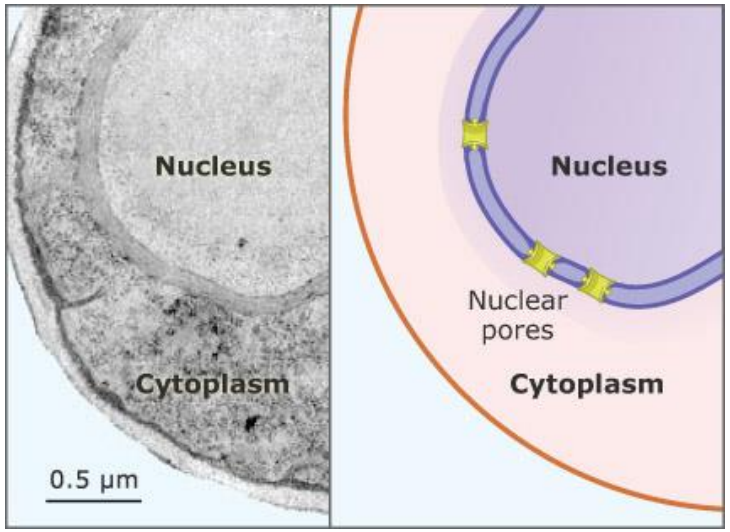


Image from Purves et al.,

Domain Eukarya/ Eukaryotic cell

- A eukaryotic cell contains many **membrane-delimited compartments**.
- Eukaryotic cells have the **nucleus** enclosed within the **nuclear membrane**.
- Within the **cytoplasm** there are individual compartments, each surrounded by a membrane.

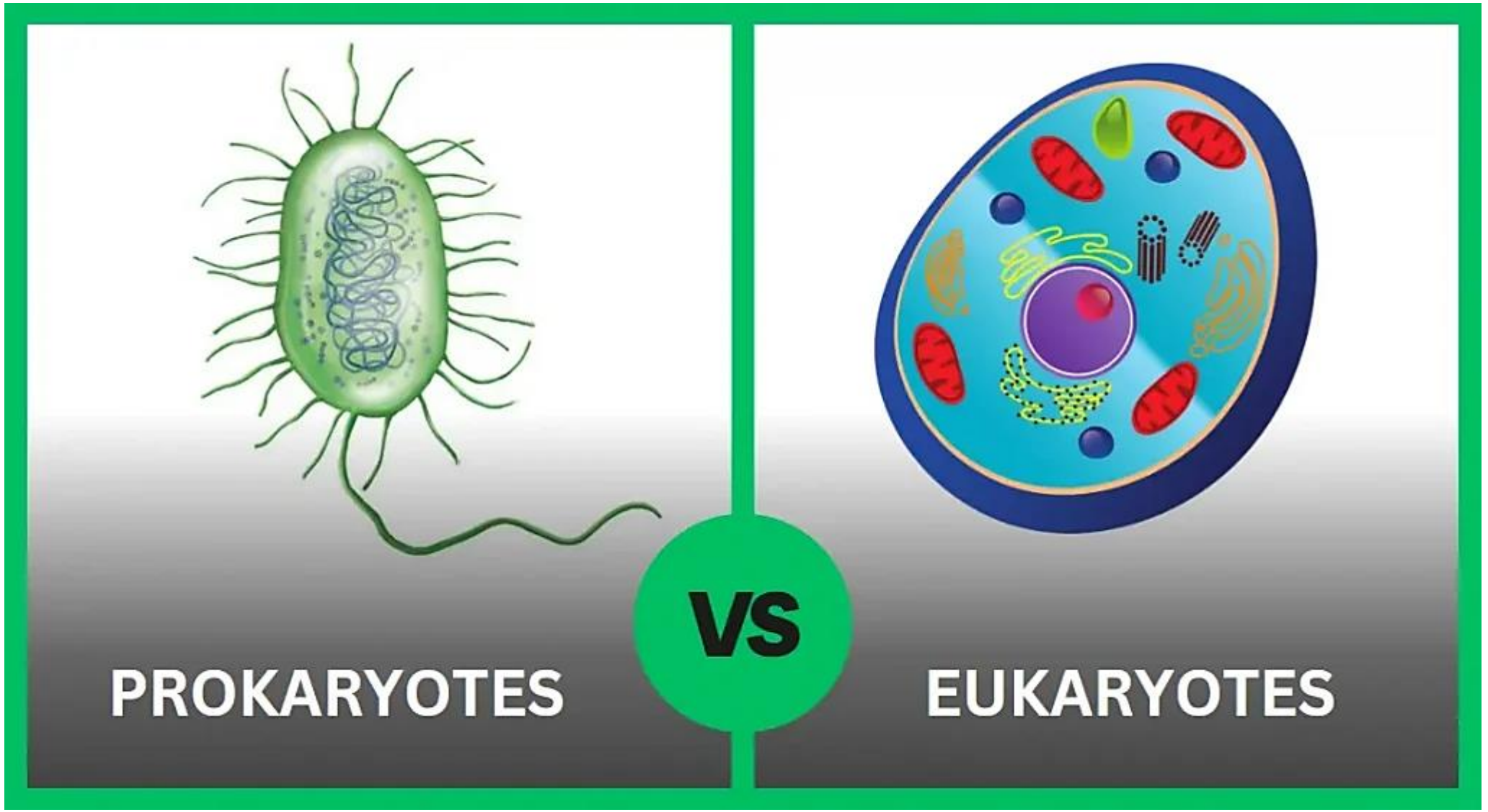


Nucleus and cytoplasm of eukaryotic cell.

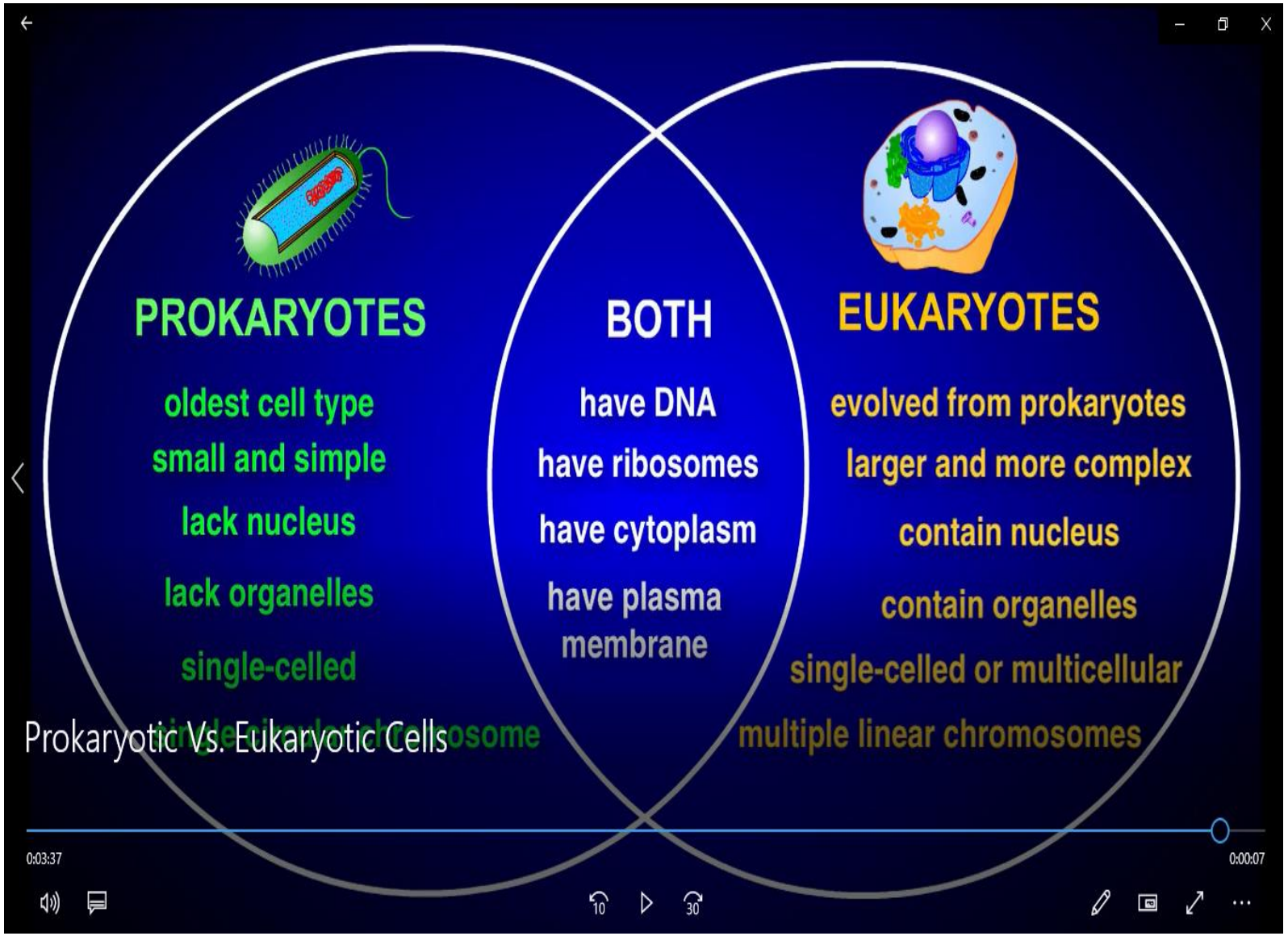
Photo © Wright, et al., 1988. Originally published in The Journal of Cell Biology, 107: 101-114. Used with permission of Rockefeller University Press. Photo courtesy of Jasper Rine, University of California, Berkeley.

Organelle	pH
Early endosome	6.5–6.8
Late endosome	5.0–6.0
Lysosome	4.5
trans-Golgi network	6.5–6.7
Mitochondria Matrix Intermembrane space	 8 7
Cytosol	7.4
Nucleus / ER	7.4

Map of cell shows different organelles have different pH values.


PROKARYOTES
VS
EUKARYOTES

Prokaryotic vs. Eukaryotic Cells


PROKARYOTES
oldest cell type
small and simple
lack nucleus
lack organelles
single-celled
BOTH
have DNA
have ribosomes
have cytoplasm
have plasma membrane
EUKARYOTES
evolved from prokaryotes
larger and more complex
contain nucleus
contain organelles
single-celled or multicellular
multiple linear chromosomes
Prokaryotic Vs. Eukaryotic Cells
0:03:37
0:00:07

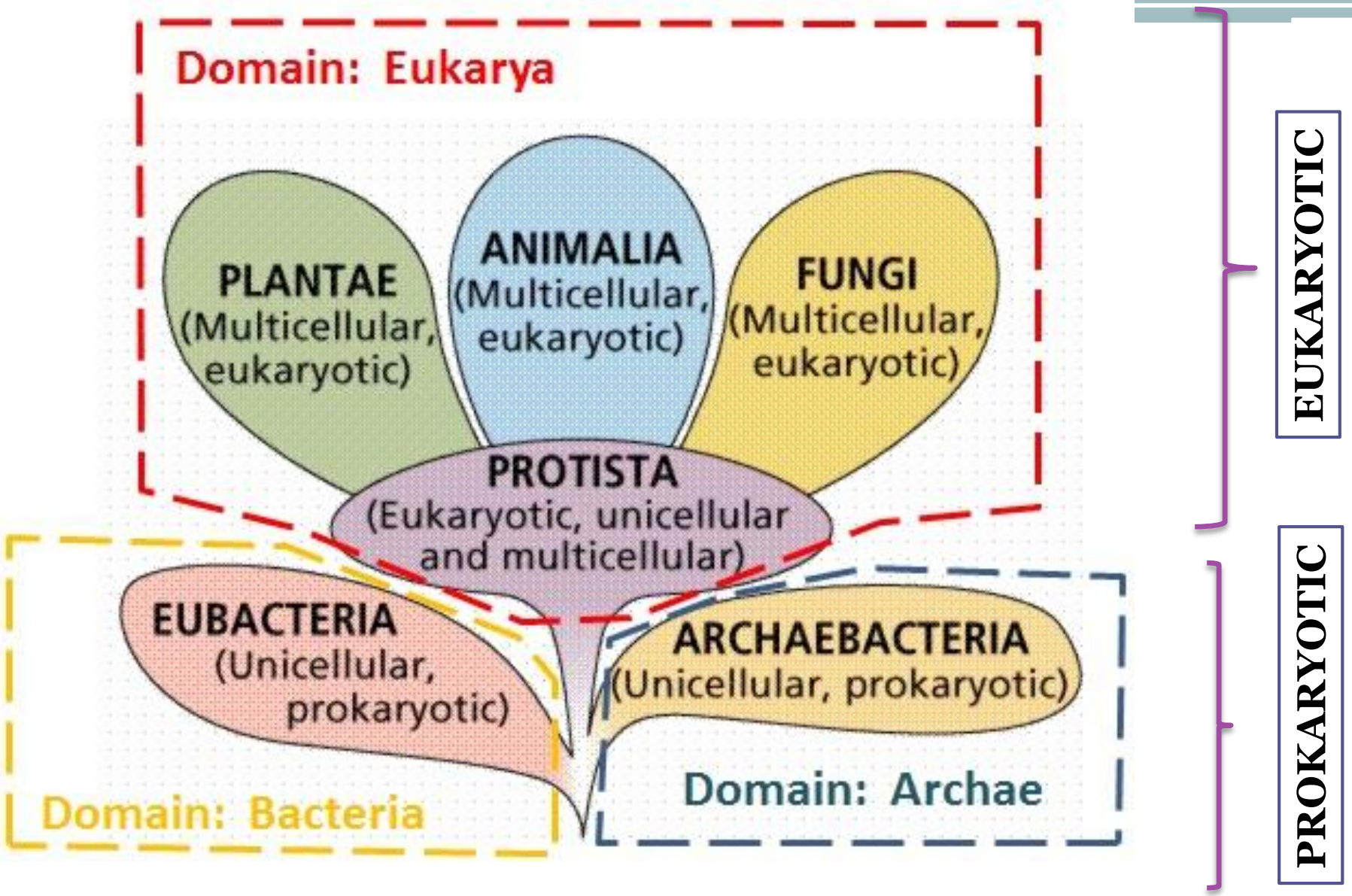


Image from Purves et al.,

Protista

Composed of a single cell or a colony of similar cells

Live in water, in moist terrestrial habitats, and as parasites and other symbionts in the bodies of multicellular eukaryotes. **Autotrophs, heterotrophs, or both.**

Reproduce asexually, some sexually

They are **best defined as eukaryotes that are NOT fungi, animals, or plants.** They are **polyphyletic**



Euglena



Kelp



Trypanosoma



Slime molds

Comatricha nigra with developing fruiting bodies.

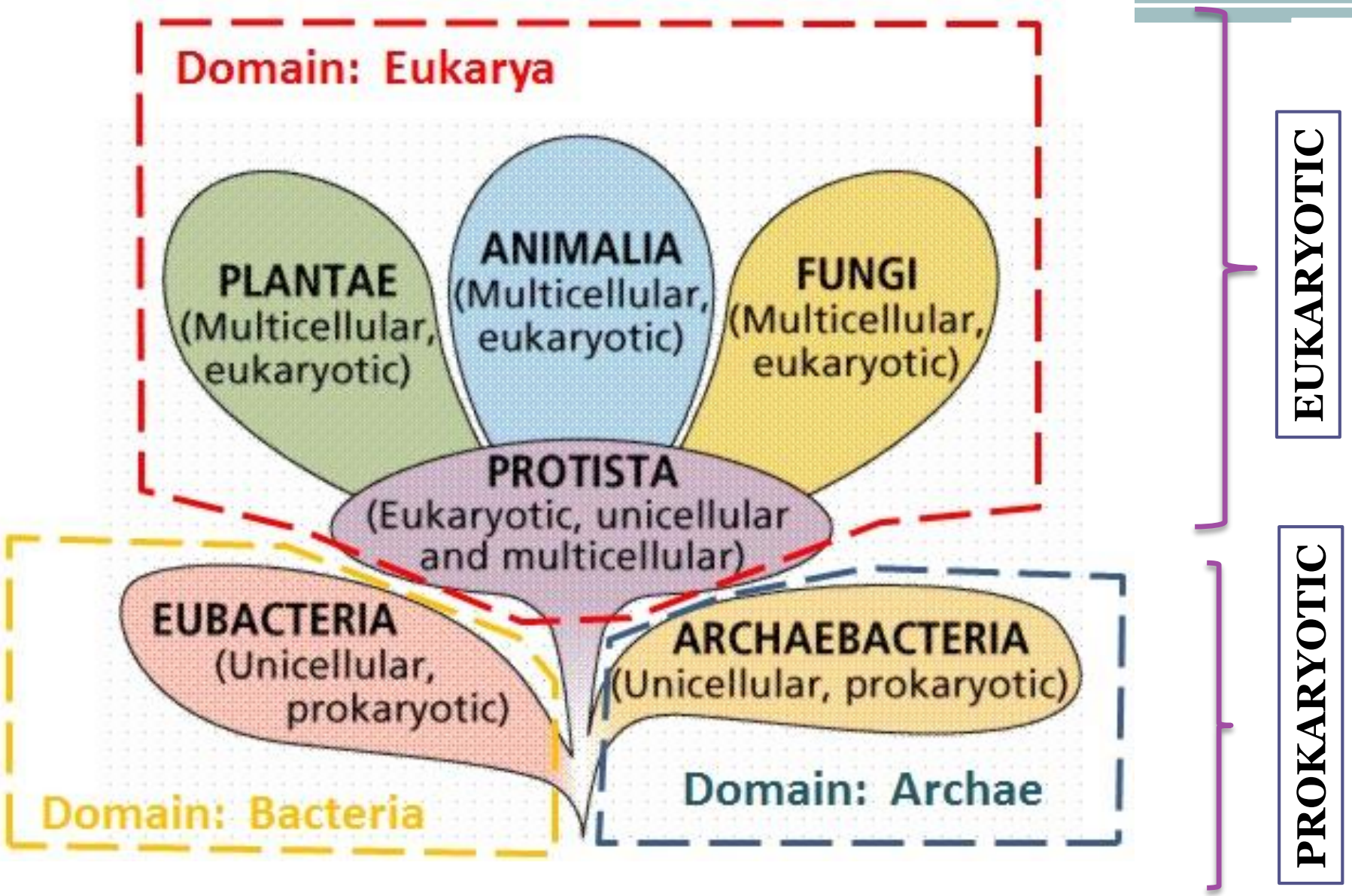


Image from Purves et al.,

Fungi

Eukaryotic.

Decomposers – the best recyclers around.

No chlorophyll – non-photosynthetic.

Most multicellular (hyphae) – some unicellular (yeast)

Non-motile.

Cell walls made of **chitin**

Are more related to animals than the plant kingdom.







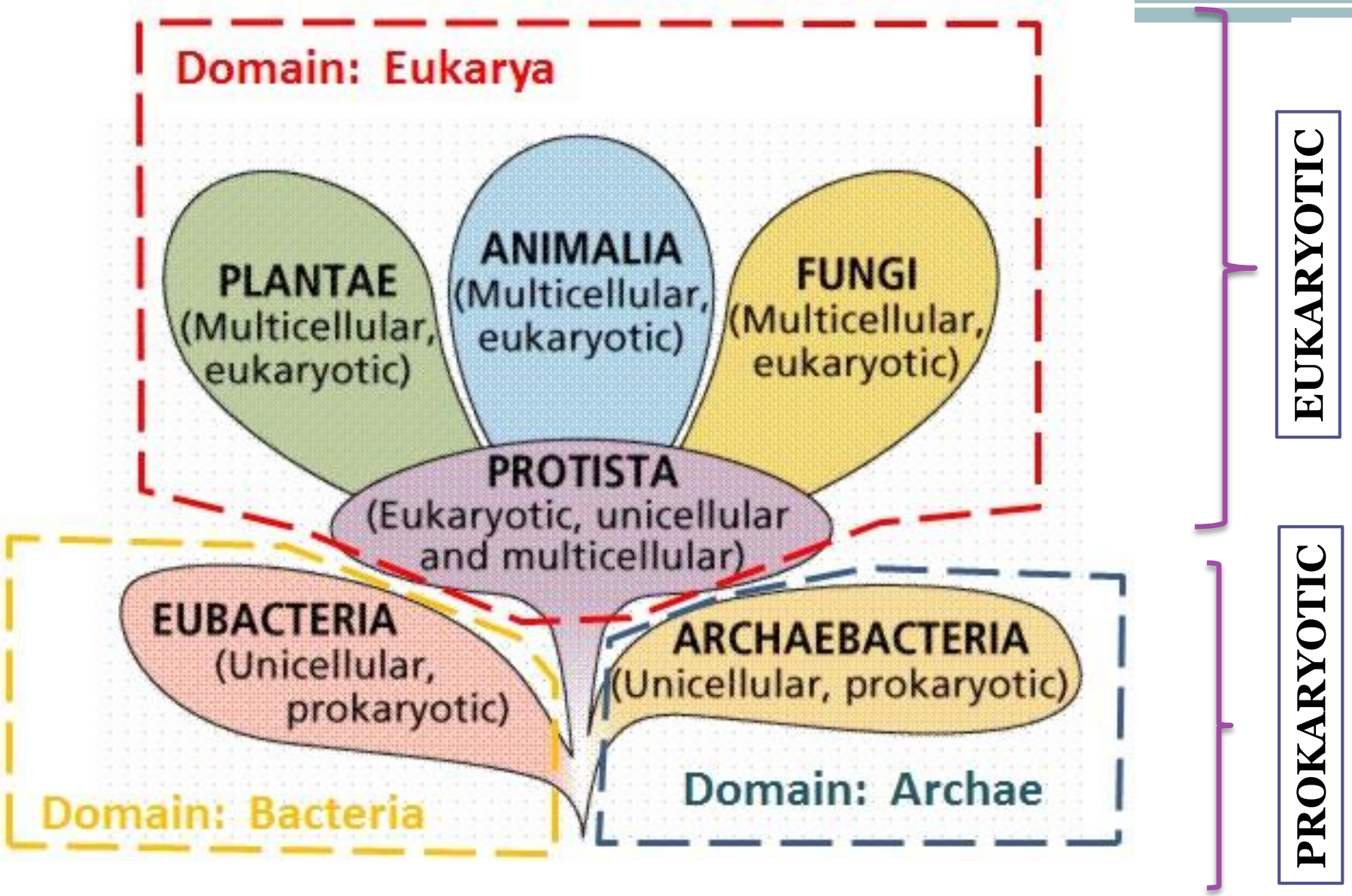


Image from Purves et al.,

Plantae

Multicellular, **eukaryotes** and consist of a rigid structure that surrounds the cell membrane called the **cell wall** (made of cellulose)



The plant kingdom has the following characteristic features:

They are non-motile

They usually reproduce sexually

They follow the **autotrophic mode** of nutrition

These contain photosynthetic pigments called chlorophyll in the plastids.

Classification of Kingdom Plantae

Plant body: whether the body has well-differentiated structures or not.

Vascular system: whether the plant has a vascular system for the transportation of substances or not

Seed formation: whether the plant bears flowers and seeds or not; if it does, then whether it is enclosed within fruits or not.







Classification of Kingdom Plantae

All the plants that lack a well-differentiated body structure belong to the subgroup Thallophyta



Thallophytes commonly include members with primitive and simple body designs such as green algae and brown algae. The majority of them are aquatic

https://byjus.com/biology/plant-kingdom-plantae/

Classification of Kingdom Plantae

Small, non-vascular plants that prefer moist environments





Bryophytes have differentiated plant body like stem, leaf structures. But they lack a vascular system for the transportation of substances across the plant body

https://byjus.com/biology/plant-kingdom-plantae/

Classification of Kingdom Plantae

Pteridophytes have well-differentiated structures such as stem, root, leaves as well as a vascular system



Pteridophytes: Spore-dispersing vascular plants

https://byjus.com/biology/plant-kingdom-plantae/

Classification of Kingdom Plantae

Gymnosperms are plants that have well-differentiated plant body, vascular system and they bear seeds

gymno: naked and *sperma*: seed



Vascular plants that possess “exposed” seeds

https://byjus.com/biology/plant-kingdom-plantae/

Classification of Kingdom Plantae

Angiosperms are also seed-bearing plants with well-differentiated plant body

angio: covered and *sperma*: seed

Angiosperms are commonly known as flowering plants



Vascular plants that possess special characteristics such as flowers and fruits

https://byjus.com/biology/plant-kingdom-plantae/

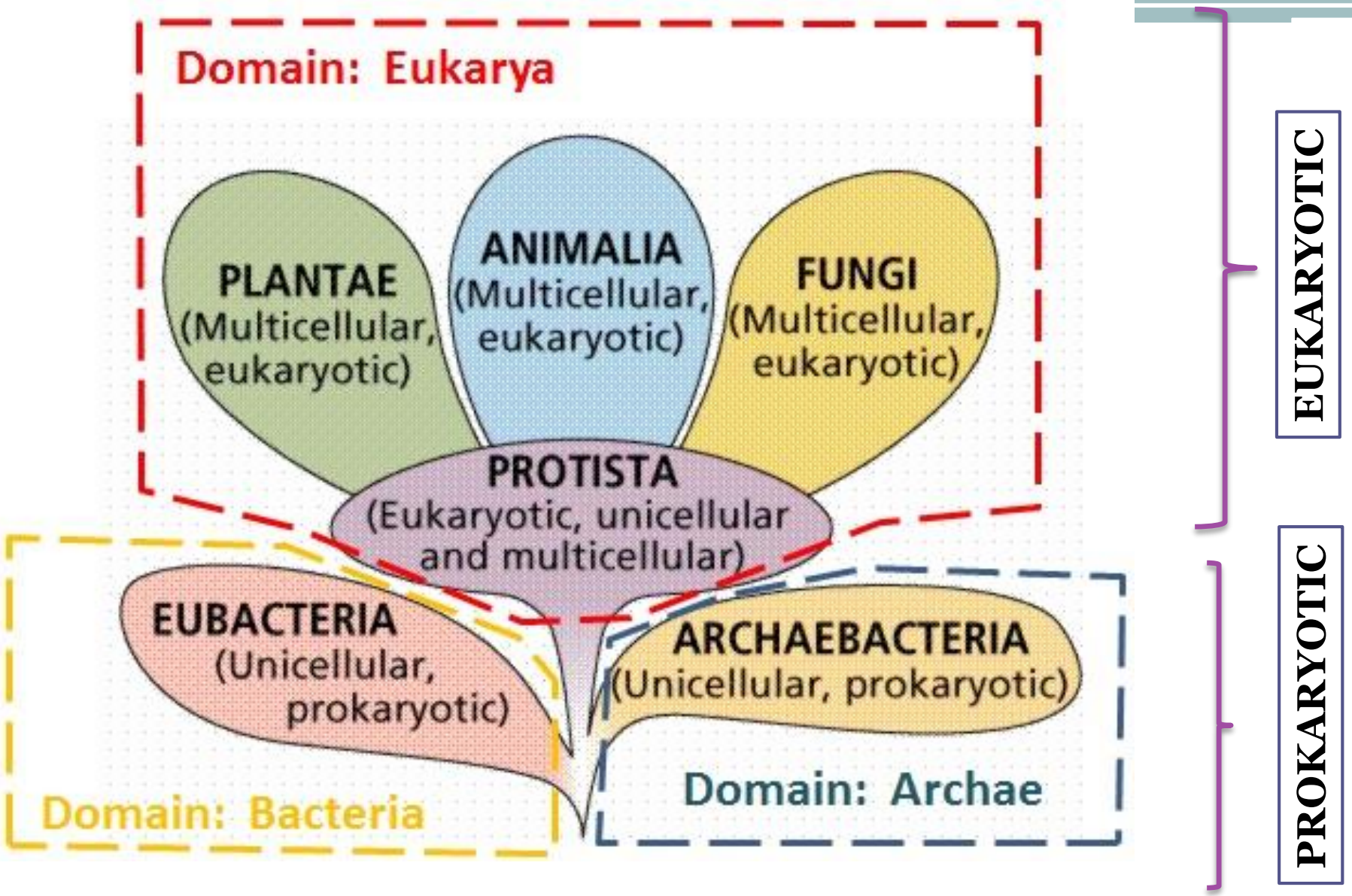
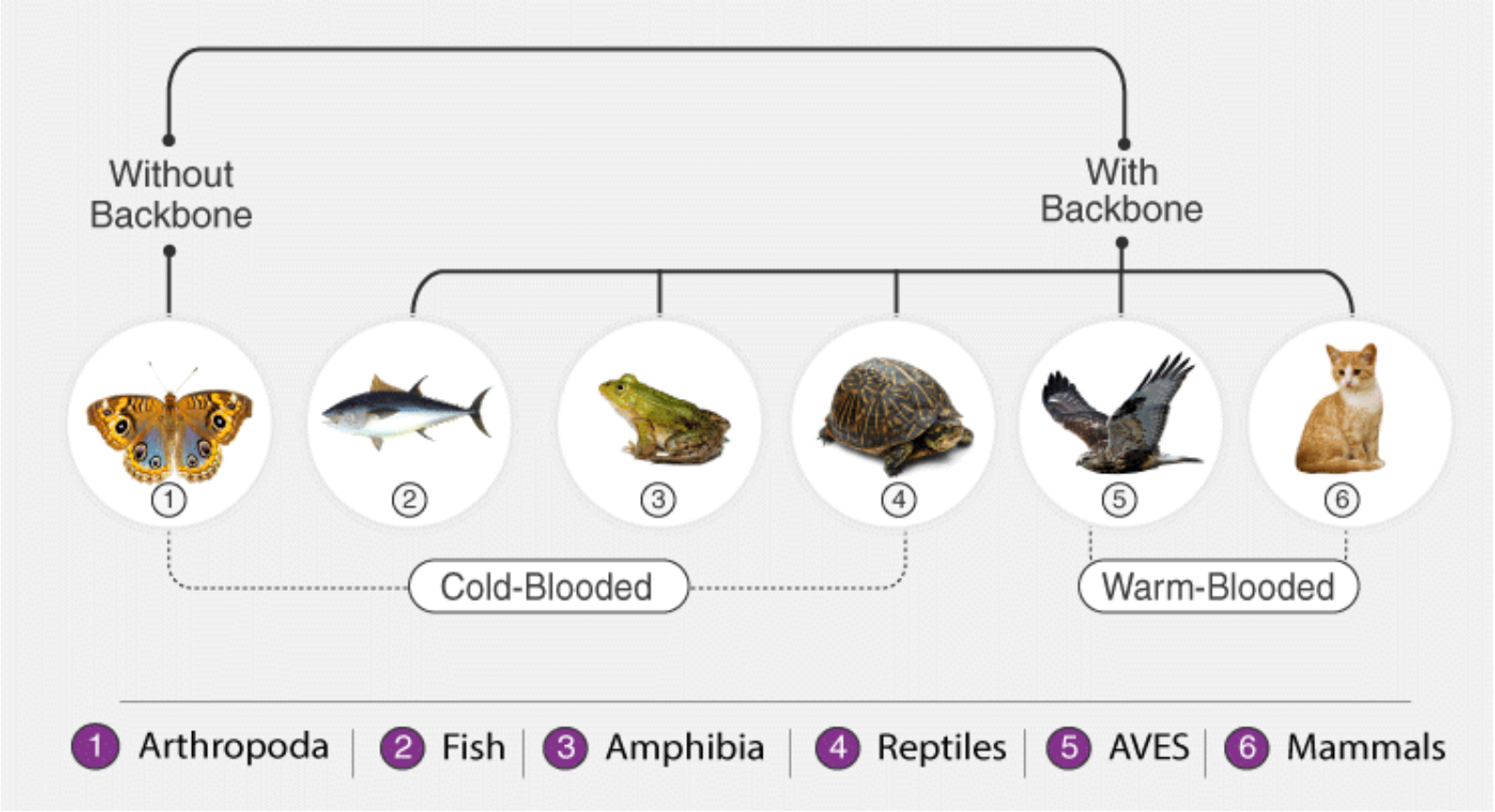


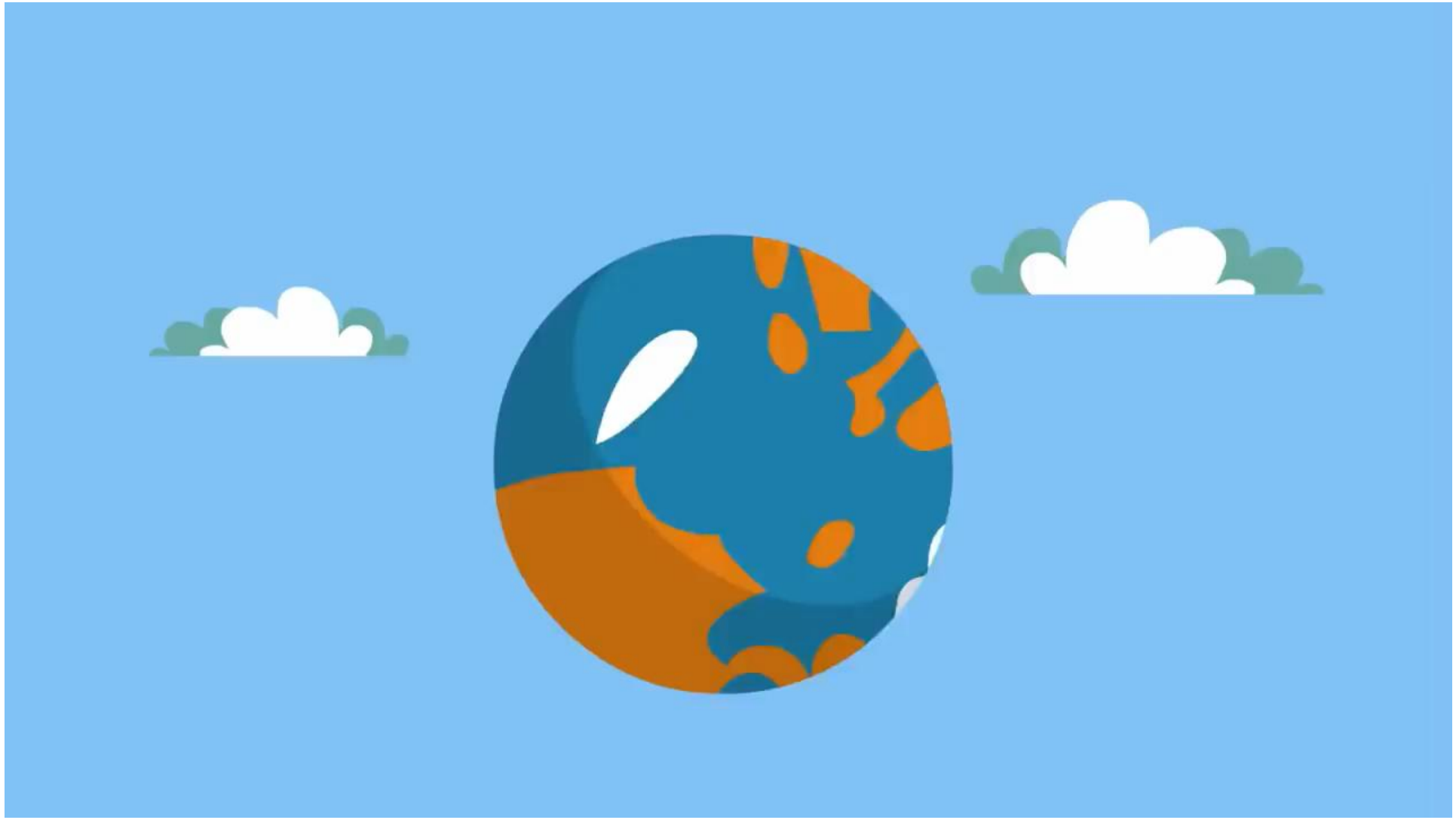
Image from Purves et al.,

Animalia

Multicellular, heterotrophic eukaryotes that are capable of **mobility** at least for part of their life, and that have cells lacking cell walls

Classification





Take home messages

Cells are the fundamental units of life.

2 types of organism: Unicellular and Multicellular

Cell difference in size, shape, structure, and function

Spontaneous generation does not occur

2 cell types: Eukaryotic and Prokaryotic/ common and difference?

3 domains, 5 kingdoms and characteristics

Your issues/questions/ comments/suggestions for us



https://forms.gle/1uv2s8LX8AFzRbis8